

Tariq Akilah

San Jose, CA | US Citizen | (408) 333-4327 | tarakilah101@gmail.com | linkedin.com/in/tariq-akilah | www.tariqakilah.com

EDUCATION

Cumulative GPA: 3.88

University of California, Irvine

Bachelor of Science, Mechanical Engineering

Irvine, CA

Expected 12/2028

Mission College

Associate Degree for Transfer, Physics

Santa Clara, CA

06/2026

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360, Onshape, AutoCAD, GD&T / Tolerance Analysis, Engineering Documentation, Engineering Drawings

Simulation & Analysis: STAR-CCM+ (CFD), SimScale (CFD), OpenRocket

Manufacturing & Fabrication: FDM 3D Printing, MIG Welding, Soldering, Mechanical Assembly

Programming: MATLAB (Simulink), C / C++, Python

EXPERIENCE

Aerodynamics Engineer

07/2026 – Present

Anteater Electric Racing - FSAE

Irvine, CA

- Increased undertray downforce by 200% by diagnosing turbulent inflow off the front wing's trailing edge, reprofiling the undertray inlet geometry in SolidWorks, and validating with CFD in STAR-CCM+.
- Coordinating with the front wing, rear wing, and sidepod subteams to ensure the undertray performs as part of one cohesive aero package, and working with the suspension subteam to finalize an operating ride height.

Fluid Dynamics Research Intern

06/2026 – 08/2026

Mission College

Santa Clara, CA

- Validated CFD mesh convergence in SimScale for [Thrust Stack](#), a research team developing an active roll-stabilizing rocket through canard control, refining mesh cell count to produce consistent roll torque coefficient results within 5%.
- Delivered a 162-point roll torque dataset adopted into the rocket's flight control logic, selecting a smaller canard geometry for finer control, sweeping deflection from 0.55 to 9.9 degrees and freestream velocity from 33 to 113 m/s.
- Designed and fabricated a custom servo housing in SolidWorks, integrating it flush with the airframe to cut the drag coefficient by 26%.

Avionics Lead

07/2025 – 06/2026

Mission Launch Rocketry

Santa Clara, CA

- Reduced avionics bay mass by 34% for a simulated 318 ft apogee gain by redesigning its layout, components, and material selection, and tuning FDM parameters.
- Modeled flight in OpenRocket, achieving a 1.12-cal stable launch and predicting apogee within 7% of previously measured flight data.
- Integrated dual-redundant flight computers that recovered critical data from two failed launches, diagnosing a faulty barometric reading and wind-induced tilt as root causes, then widened angle tolerance to 45° for a final flight simulated at 5,400 ft.

PROJECTS

NAR High Power Rocket Certification | Fusion 360, OpenRocket, FDM

- Designed and manufactured a high-power rocket in Fusion 360, FDM-printing the airframe from PETG-CF on a Bambu Lab P1S to achieve geometric symmetry under OpenRocket simulated flight loads while staying under 600 grams.
- Executed a stable Mach 0.747 flight to 4,600 ft, recovering the airframe without critical damage to earn NAR Level 1 High-Power Certification.
- Passed a pre- and post-flight inspection from a NAR Level 3 mentor, verifying structural rigidity under load, proper construction and assembly, and motor retention and tolerancing.

Motorized Trike | Fusion 360, MIG Welding

- Built a motorized trike from recycled materials on a \$500 budget and a 5-week program deadline.
- Modeled the chassis in Fusion 360 and fabricated it from 1.75 mm wall steel square tubing using self-taught MIG welding, validating structural rigidity under a 180 lb rider and lateral drifting loads with minimal chassis flex.
- Rebuilt a seized 212cc engine and mated a go-kart rear axle to a bicycle front fork, reaching a 40 mph top speed.

Dissolved Air Flotation (DAF) Tank Redesign | AutoCAD, SolidWorks, Engineering Documentation

- Designed a one-to-one replacement DAF tank in SolidWorks for the San Jose-Santa Clara Wastewater Facility, matching existing plant dimensions and interfaces to drop into the current footprint with no changes to surrounding infrastructure.
- Delivered a complete engineering package with SolidWorks models, AutoCAD civil/electrical site plans, and industry-standard contract documentation, reviewed and approved by a former facility engineer, earning full marks.

LEADERSHIP AND ACTIVITIES

Co-Founder & Vice President

07/2025 – 06/2026

Mission Launch Rocketry

- Grew the first Mission College project-based engineering club to 60+ members, with 14 active on its first project, raising \$5,300 in funding.

Founder & President

09/2025 – 06/2026

Mission College Muslim Engineering Society

- Founded a Muslim STEM networking society, growing to 35+ members and landing internships for 2 through engineering speaker events.

Secretary

03/2025 – 06/2026

Mission College Associated Student Government

- Published 70+ official agendas and meeting minutes, coordinating communication across students, faculty, and administrators.